

90-Minute 4K HLS Movie: QueryBook UFCS-FQL vs. Standard Delivery

Oct 5, 2026 · @Garry Howard

The short answer

For one 90-minute 4K movie in HLS, QueryBook UFCS-FQL and standard delivery need the same storage (about 9 GB at 12 Mbps H.265) and give the same picture. Standard web delivery moved the movie about 12% faster. QueryBook adds a fingerprint check on every piece and proves the whole movie arrived identical.

	Standard HLS (web delivery)	QueryBook UFCS-FQL + TCP/IP
Storage, whole movie	8.95 GB (modern HLS)	8.95 GB (same file; wrapping adds 0.3% only while moving)
Picture quality	SSIM 0.848	SSIM 0.848 (same encode)
Time to move the movie over a 50 Mbps line	24.5 min	27.9 min
Data on the wire	8.95 GB	8.98 GB
Every piece checked and whole movie proven identical	Not by default	Yes

The biggest saving does not come from the delivery method. It comes from the video format: AV1 made the movie 14% smaller than H.265 at the same quality in our test.

Storage

Modern HLS (MP4 pieces) is the leanest way to store the movie; QueryBook stores that same file. Old-style HLS (TS pieces) costs 2.4% more.

How the 4K top rung is stored (H.265, 12 Mbps)	Whole 90-min movie	vs leanest
Modern HLS, MP4 pieces	8.95 GB	leanest

How the 4K top rung is stored (H.265, 12 Mbps)	Whole 90-min movie	vs leanest
QueryBook UFCS-FQL frames (only while travelling)	8.98 GB	+0.3%
Classic HLS, TS pieces	9.16 GB	+2.4%

A full HLS quality ladder (4K down to phone sizes) typically adds another 40 to 60% on top of the 4K rung, the same for both approaches.

Compression: which video format

Both approaches can use any of these formats; this is the choice that moves the size.

Format at 12 Mbps target	Whole 90-min movie	Picture match (SSIM, 1.0 = perfect)
AV1	7.69 GB	0.849
H.264	8.61 GB	0.843
H.265	8.95 GB	0.848

AV1 gave the best picture in the least space. The test footage is deliberately grainy, which is hard for every format, so quality scores are lower than for a typical clean movie.

UFCS-FQL's own compression vs. standard video codecs

Standard video codecs win by a huge margin: H.265 made real-looking video 105 times smaller, while UFCS-FQL's own compressors managed less than 2 times. That is why UFCS-FQL hands video to H.264, H.265 or AV1 and keeps its own compressors for data and text.

The reason: UFCS-FQL's compressors (Zstd, GZIP and the UFCS dictionary) are *lossless*. They keep every pixel exactly, like zipping a file. Video codecs are *lossy*: they throw away detail the eye won't miss, which is where the big savings come from.

Measured on a 2-second 1080p clip (149 MB uncompressed), scaled to a 90-minute movie:

Compressor	Type	Times smaller, real-looking (grainy) video	90-min movie size	Times smaller, clean graphics	Picture match (SSIM, 1.0 = perfect)
None (raw)	—	1×	403 GB	1×	1.000
UFCS-FQL GZIP	Lossless	1.7×	238 GB	21×	1.000
UFCS-FQL dictionary	Lossless	1.7×	238 GB	26×	1.000
UFCS-FQL Zstd	Lossless	1.8×	221 GB	32×	1.000
FFV1 (standard lossless video codec)	Lossless	2.7×	149 GB	29×	1.000
H.264 at 6 Mbps	Lossy	91×	4.4 GB	90×	0.952
AV1 at 6 Mbps	Lossy	99×	4.1 GB	75×	0.956
H.265 at 6 Mbps	Lossy	105×	3.8 GB	98×	0.955

What this means in plain terms:

- **For real video, use the standard codecs.** Lossless squeezing of a 90-minute film still leaves over 200 GB; H.265 gets it under 4 GB with a picture that looks nearly identical.
- **For an archive master that must keep every pixel,** UFCS-FQL Zstd is a fair choice but the standard FFV1 codec is 1.5 times smaller.
- **On simple graphics** (slides, cartoons, screen recordings), UFCS-FQL Zstd beats FFV1 (32× vs 29×) while keeping every pixel exact.
- **Re-squeezing already-encoded video does nothing.** Running UFCS-FQL compressors over the 4K H.265 movie file saved 0.0%, so UFCS-FQL sends codec output untouched.

All UFCS-FQL results were checked lossless: unpacked, each matched the original byte for byte.

Delivery: moving the whole movie

On a 50 Mbps line with 20 ms delay each way, standard web delivery was about 12% faster. A viewer watching it stream needs 12 Mbps either way.

	Standard web download	QueryBook UFCS-FQL
Time for a 6-second 4K piece (measured, 2 runs)	1.63 s	1.86 s

	Standard web download	QueryBook UFCS-FQL
Whole 90-min movie (scaled)	24.5 min	27.9 min
Bytes sent	8.95 GB	8.98 GB

Why QueryBook is slower here: it uses three lanes and waits for a signed receipt, and the lab nodes are written in PHP. Its advantage appears on busy lines carrying live video, where urgent pieces skip the queue (0.14 s vs 0.60 s in our earlier live test). A single on-demand movie doesn't benefit.

Accuracy and reliability

The picture is identical, because the same encode is delivered both ways. QueryBook adds proof that it arrived intact.

Check	Standard HLS	QueryBook UFCS-FQL
Each piece fingerprinted on arrival	Network's basic check only (stronger with HTTPS)	Yes, every piece
Whole movie proven identical	No	Yes (SHA-256 match confirmed in the test)
A damaged piece	Player may stall	Piece dropped, rest continues

Cost at scale

Per 1,000 full viewings, the delivery method changes the bill by well under 1%; the video format changes it by 14%. Assumes an illustrative CDN price of \$0.02 per GB.

Choice, per 1,000 full viewings	Data sent	Cost at \$0.02/GB
Modern HLS, H.265 (baseline)	8.95 TB	\$179
QueryBook UFCS-FQL to every viewer	8.98 TB	\$180 (+\$0.50)
Classic HLS, TS pieces	9.16 TB	\$183 (+\$4)

Choice, per 1,000 full viewings	Data sent	Cost at \$0.02/GB
Modern HLS, AV1	7.69 TB	\$154 (–\$25)

Best setup: AV1 in modern HLS to viewers; QueryBook UFCS-FQL between your own sites, where its checks and priority lanes pay off.

How we tested

- A 6-second lossless 4K (3840×2160 , 24 fps) grainy test clip, encoded at a 12 Mbps target with H.264, H.265 and AV1, then packaged as HLS with no re-encoding. Results scaled $\times 900$ to 90 minutes.
- Delivery timed over a simulated 50 Mbps, 20 ms line on one machine, 2 runs each; runs agreed within 0.05 s.
- The \$0.02/GB price and the 40–60% ladder figure are typical industry estimates, not measured. Real films, real networks and a full ladder will shift the numbers, not the conclusions.